REMORA: Real-time Expressive Motion to Output Routing Audio

A Martial Arts Staff Instrument Driven by Motion

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ABSTRACT

REMORA turns a bô, the long staff used in martial arts, into a musical instrument: spins, tilts, and strikes become sound in real time. The instrument is built in two parts: a small wireless sensor unit rides on the staff and continuously reads its physical motion, while a companion audio plugin turns that motion into pitch, texture, and timbre across three very different synthesis voices, from a warm additive choir to a breath-driven vocal tract. Rather than one fixed gesture-to-sound behaviour, every mapping from physical motion to sound parameters is a small patch of connected building blocks - motion sources, shaping math, and synthesis sinks - wired on a visual canvas inside the plugin, in the spirit of a modular synthesiser's patch cables. The same small vocabulary of building blocks stretches from a beginner's direct tilt-controlled pitch to a compound, deliberately one-to-many mapping whose timbre quietly re-randomises itself with every completed spin, so that no two passes through the same gesture sound quite identical. Nine such presets currently ship, pared down over the course of development from a larger, more repetitive set of fifteen.

1. INTRODUCTION

REMORA grew out of the first author's parallel practice in martial arts and music. The bô itself is a form largely absent from new interfaces for musical expression: a full-body, two-handed, swung staff with its own weight, reach, and rotational inertia, quite unlike the small handheld or body-worn sensors most inertial digital instruments use. Where a martial artist reads speed, force, and reach off the staff by feel, REMORA reads the same qualities electronically and turns them into sound: a spin becomes a filter sweep, a slow controlled turn sounds nothing like a fast, loose one. The instrument's vocabulary is, in that sense, borrowed directly from the performer's own body of practice rather than invented for the occasion. Crucially, that vocabulary is not baked into compiled code either: every motion-to-sound relationship lives inside the plugin as a small, rewireable patch, so extending or retuning it is an editing operation the performer or instrument-builder can perform on the same canvas used to play the instrument, rather than a recompile.

Rod-, staff-, and baton-shaped controllers already form a small, recurring family in this space: the T-Stick's touch-sensitive shaft [1], the Digital Baton's conducting gesture [2], and Nam's swung Musical Poi [3] among them. What REMORA adds to that family is less a new sensor than a new relationship: a martial artist's years of staff-handling skill, carried straight onto a stage as a musical vocabulary. The remainder of this paper describes how that relationship is built, from the sensor unit on the staff (§2) through the motion-to-sound mapping layer (§3) to the resulting catalog of presets (§4).

2. SYSTEM DESIGN

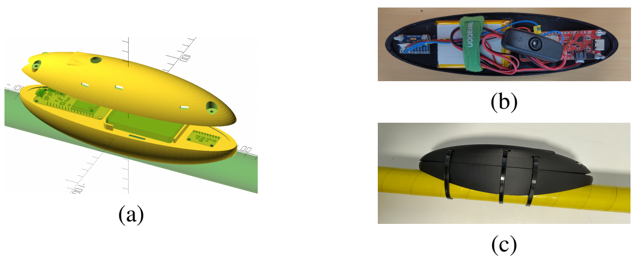

Figure 1: The staff-mounted sensor unit, held on with zip ties so it can be repositioned or removed without marking the staff: (a) the parametric OpenSCAD design; (b) the harware wiring; (c) the printed enclosure zip-tied to the bô.

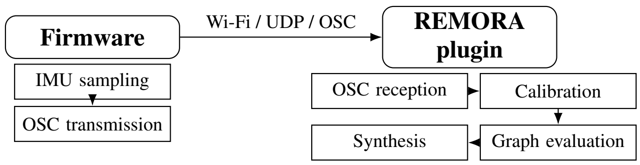

Figure 2: Signal path from staff motion to sound: the sensor unit streams orientation and inertial data over Wi-Fi to the plugin, which calibrates, derives motion qualities, and routes them through a performer-authored patch into one of three synthesis engines.

2.1. Hardware

Inside the enclosure shown in Figure 1, a SparkFun ESP32-S2 Thing Plus microcontroller reads a nine-degrees-of-freedom MPU-9250 inertial sensor over I2C and packs orientation, acceleration, and gyroscope readings into an OSC packet, sent over UDP at 100 Hz across the microcontroller's own connection to the host computer's Wi-Fi hotspot. Joining the host's own hotspot rather than a dedicated access point keeps pairing to a single step, at the cost of a Wi-Fi network name, password, and destination address that are presently compiled into the firmware rather than set at runtime. Nothing tethers the performer to a fixed point on stage; a cable is needed only to charge the unit between sets.

2.2. Calibration

Because the sensor can be glued to the staff at any angle (exact placement is never perfectly reproducible by hand, and often changes from one build of the enclosure to the next), the plugin calibrates itself the first time the staff is held through three reference poses. All three poses together fix which physical axis is functioning as "forward": the plugin scores every candidate axis by how close to mutually orthogonal its direction turns out across the three poses, and keeps whichever axis comes closest. The first two poses then resolve any leftover twist around that axis, by solving for the small rotation that best lines up the measured forward/up directions with the instrument's own ideal frame. From then on, the same instrument logic runs correctly no matter where on the staff the sensor happens to sit, so a performer can move it without re-learning the instrument underneath.

2.3. Connection Robustness

Two recent changes tightened the path from a sensor packet to audible sound. First, the plugin now treats the incoming stream as stale after 150 ms without a fresh packet, rather than the previous 500 ms, so a real Wi-Fi dropout is caught and muted noticeably sooner. Second, that mute no longer shares a single ramp with normal playing: losing the signal now fades the master gain out over 120 ms (long enough to avoid an audible click) while recovering it fades back in over a much shorter 10 ms, so the instrument goes quiet gracefully on dropout but comes back almost immediately once the connection returns.

3. MAPPING DESIGN

3.1. Motion Qualities

Every packet from the sensor is read against a moving reference: not just where the staff is, but how it got there. Four continuous qualities borrowed from Laban Movement Analysis are extracted from the motion stream every audio block. How forcefully the staff moves becomes Weight, read from a leaky integral of its acceleration once gravity is subtracted out, and typically driving loudness or waveshaping. How suddenly a spin speeds up or stops becomes Time, driving attack speed and how quickly a chord change lands. Whether the staff keeps spinning around the same axis or wanders becomes Space, a timbral focus that tightens or loosens a filter or a chorus spread. How bound or free the motion is becomes Flow, crossfading between a controlled and a freer, noisier version of the same sound. Tilt and roll bend pitch and blend voices continuously; a spin's lap count sweeps a filter cutoff along a slow curve; a stab or strike, a sharp reversal in the staff-tip's acceleration, triggers a transient, a chord change, or a percussive burst.

3.2. Visual Patching

None of the relationships described above is a single fixed algorithm baked into compiled code. Every one of them is authored as a small patch of connected building blocks inside the plugin, on a visual canvas in the lineage of Max [4] and Pure Data [5], closer in feel to the boxes-and-cables idiom that community modular-synthesis software such as VCV Rack [6] carries over from hardware Eurorack patching than to a conventional plugin's fixed control panel. Three functional kinds of node populate that canvas: *sources* (the raw sensor axes and the derived Laban qualities above), *math* (arithmetic; shaping curves such as range-mapping, logarithmic range-mapping, clamping, crossfading, and scale-quantizing; lookup tables; and a handful of stateful primitives - one-pole smoothers, leaky integrators, derivative detectors, hysteresis gates, and a pseudo-random sample-and-hold), and *sinks* (the exposed parameters of whichever synthesis voice the patch is driving). A fourth kind, *display*, carries no signal of its own: a numeric readout, a level meter, and a scrolling scope can be dropped inline on any wire purely to watch its live value, without altering the patch. A patch is a directed graph with no cycles allowed: the plugin topologically sorts it once whenever a wire changes, then re-walks that fixed order once per audio block, so dragging in a node or dropping a new connection mid-performance never forces the audio thread to replan anything. A named preset is nothing more than one such graph, serialised to an XML file and reloaded by name - the same mechanism a performer's own live patch is saved through, so a custom mapping built during a rehearsal is stored no differently from a factory one.

That shared vocabulary is small enough to describe in full, yet stretches from the simplest mapping in the catalog to a considerably more elaborate one. *Simple* is close to the shortest patch the graph can express: the pitch source feeds one range-mapping node that becomes the root frequency (wired straight into the additive voice's sink), and the roll source feeds an absolute-value node into a second range-mapping node that becomes the gain (wired into its own, separate gain sink) - two motion sources and three math nodes feeding two sink nodes in total, wired once and left alone for the rest of the performance. *Azimut Kinetic* sits at the other end of the same vocabulary. Its pitch and gain follow the same lookup-table-driven facing logic as the rest of the Azimut family, but its timbre is deliberately one-to-many [7]: a single smoothed rotation-speed value fans out simultaneously into the filter cutoff, harmonic brightness, waveshaping drive, noise amount, vibrato and tremolo depth, and reverb send, so a spin sounds progressively buzzier, noisier, and more reverberant the faster it turns, rather than moving one parameter in isolation. Layered on top of that energy value, a derivative node watches the continuous spin-lap counter for the instant each full rotation completes and fires a one-block pulse into a sample-and-hold node, which then latches four fresh pseudo-random values; each value is chased by its own slow one-pole "drift channel" - closing roughly 5% of the remaining distance to its new random target on every audio block, so the drift glides rather than jumps - and folded back, in a small proportion, onto the cutoff, drive, noise amount, and harmonic brightness the energy value already set. The result is a mapping that never quite repeats itself: two visually identical spins land on two different, nearby timbres, because every completed rotation quietly re-rolls where the tone drifts to next - an instability the graph's existing math nodes are enough to build, with no bespoke code and no audio-thread cost beyond four extra smoothers.

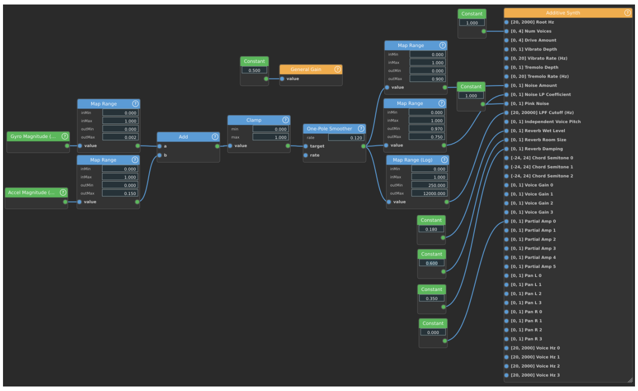

Figure 3: The *Wind Noise* preset, seen inside the plugin: motion sources (green) - gyroscope and accelerometer magnitude - run through shaping and smoothing math (blue) into the exposed parameters of the additive-synthesis voice (orange). There is no pitch input anywhere in this patch: only motion energy, shaping noise amount and the master filter cutoff.

3.3. Synthesis Engines

What a patch's output actually sounds like depends on which of three synthesis voices its sink nodes are wired into. A default additive engine sums four voices of six phase-locked harmonics each, vibrato- and chorus-detuned into a warm, faintly unstable choir that a mapping can drive into a rasping saturation or filter down to a single dark hum; a short comb resonator gives every voice a string-like colouration. That same engine also carries its own built-in noise generator and filter, and it is this path, not a separate voice, that *Wind Noise* actually drives (Figure 3): the patch wires nothing at all into the engine's pitch inputs, only motion-energy-shaped noise and filter cutoff, for a windy, textural character with no fixed pitch centre at all. A second, granular engine sits alongside the additive one, synthesising its own two-second bed of shifting harmonics and filtered noise and scattering up to 64 overlapping grains across it (forward, backward, panned, repitched); it is fully implemented but not yet wired into any preset in the current catalog, so it stays silent until a future patch calls on it. A third engine ports Neil Thapen's *Pink Trombone* [8], a digital-waveguide vocal tract excited by a glottal-pulse model and breathed into with filtered noise, so the instrument can also simply speak, or half-speak, rather than only sing - the one engine with a preset built specifically around it, *Vocal Tract*. An engine a patch never wires into is skipped entirely for that block, so presets built on one voice never pay for the other two.

4. PRESET VOCABULARY

Nine named presets currently ship with the instrument (Table 1), graded by the physical vocabulary they ask of a performer rather than uniform in difficulty. *Simple* asks only for basic tilt and roll, an easy way in that needs no martial background at all. The *Azimut* and *Martial* families reward the staff-handling skill a martial artist already carries onto a stage: they derive a facing direction and a spin quality straight from the calibrated motion stream, without ever touching a magnetometer (which a stage rig's own electronics would only confuse), so that where the staff points and how it turns becomes part of the music. *Wind Noise* trades pitch for a breathy,

Preset	Character
Simple	Pitch from tilt, volume from roll; one plain sine voice.
Wind Noise	No pitch; motion energy swells filtered noise into gusts.
Bowed Chord	Chromatic root from pitch; yaw/roll pick the chord; spin speed drives it like bow pressure.
Lead + Drone	Major-scale lead over a 4-voice drone; yaw crossfades two drone voices.
Martial Momentum	Root from spin plane/direction; glide speed set by spin momentum.
Azimut Kinetic	Facing-based root; a single rotation-speed value drives filter, drive, noise, and reverb at once, with per-rotation stochastic drift.
Speed Gate	Slow: pentatonic melody. Fast: crossfades into the Azimut facing/spin-based root and filter mapping.
Spin Voices (Scale)	Roll picks 1 of 4 scale-quantized voices; the rest hold their last pitch/gain.
Vocal Tract	Physically-modeled voice; pitch/roll/yaw shape glottal pitch and vowel; jabs add consonant-like bursts.

Table 1: The nine presets currently shipping, in load order.

filtered-noise texture sustained by nothing but motion; *Vocal Tract* pushes the same sensor stream into something closer to a voice than a synthesiser. Each preset is less a separate program than a different character the same instrument can take on, chosen for a set the way a performer chooses which piece to play next, rather than a fixed, single-piece score.

This catalog is smaller than it once was. Fifteen mappings were implemented over the course of development, but several turned out to be near-duplicates of one another once built - a plain spin-driven pentatonic melody and its Laban-shifted cousin, three separate Azimut variants that each added one extra timbral trick (a spin-count filter sweep, a speed-tracking filter, a motion-driven reverb) on top of an otherwise identical mapping. Rather than ship all fifteen, the list was pruned to nine: *Azimut Kinetic* folds every one of those three Azimut timbral tricks into a single one-to-many mapping (§3) instead of shipping them as separate, thinner presets, and similar consolidations retired the redundant spin- and momentum-only mappings. The six retired mappings still exist as graphs in the source tree - useful engineering reference points and starting patches to fork from - but are no longer surfaced in the plugin's own preset selector.

5. EVALUATION

Ten performers with no prior exposure to REMORA (N=10) tried the instrument in a short, informal pilot session: each was calibrated (§2.2), given a brief demonstration, then played freely across a subset of the nine presets before completing a short questionnaire. Their backgrounds were, by design, uneven: self-rated martial-arts or staff-handling experience skewed low (nine of ten at 1 or 2 out of 5), general music experience moderate, and prior experience with motion-controlled instruments mixed - a pool weighted toward newcomers to the staff form itself, the audience *Simple* was designed to admit (§4).

Six statements about the playing experience, each rated on a five-point scale, are summarised in Table 2. Comfort scored highest and most consistently: the enclosure and staff form factor did not get in the way of play. Novelty and absorption in the music rather than the mechanics ("flow") followed closely, suggesting the motion-to-sound vocabulary reads as expressive enough to disappear behind, even in a short first session. Predictability of the mapping and felt expressiveness scored lower and more variably, consistent with the free-text feedback below: several participants could point to specific gestures whose effect on the sound they could not reliably predict.

Statement	Mean (SD)
Predictability	3.4 (1.2)
Comfort	4.0 (1.0)
Novelty	3.9 (0.9)
Expressiveness	3.3 (1.2)
Absorption	3.9 (0.9)
Musical satisfaction	3.6 (1.3)

Table 2: Mean (SD) agreement, 1–5 scale, N=10, on six playing-experience statements from the post-session questionnaire.

Preference across presets split unevenly rather than converging on one: *Azimut Kinetic* was the most-picked (3/10), followed by *Wind Noise* and *Lead + Drone* (2/10 each), with *Spin Voices* and *Vocal Tract* picked once each and one participant declining to single one out ("all of them are great"). The reasons given tracked each preset's intended character: *Azimut Kinetic* for matching lived martial-arts habit ("It matches best what I know from my martial art experience"), *Wind Noise* for needing no music or martial background at all ("just pure instinct and simple movements"), *Lead + Drone* for being the easiest to steer a melody with while still sounding grounded.

The clearest actionable finding was not something we asked about directly. Three of ten participants raised the same limitation unprompted, in three different free-text answers: a striking or thrusting motion that felt physically decisive registered no equivalent change in the sound ("Occasional frustration when a movement that felt impactful had no impact on the sound"; "The thrust are not well detected"; "Add *thrust* detection support or fix the existing"). None of the nine shipping presets currently derives a dedicated signal from a stab or strike beyond the generic transient described in §3.1, so this is a concrete, convergent request for a motion quality the mapping vocabulary does not yet expose. A second, smaller cluster of feedback concerned the node-graph editor itself rather than any preset: one participant who otherwise rated the instrument highly asked for the wiring interface to be made easier, "maybe some kind of tutorial", echoing the same complexity the graph exposes to a performer who only wants to play, already named as a limitation in §6. Reported frustration was otherwise mostly framed as ordinary friction rather than a reason to stop - "no, really easy and when a bit hard it's just a learning experience!", "did not make me want to stop but had to adjust" - with one exception noting the instrument felt enjoyable to play but did not, on its own, motivate further practice.

6. DISCUSSION

Everything described above is grounded in the system as it runs today, exercised continuously through the first author's own development and rehearsal, and now also in the pilot session reported in §5: ten first-time players found the instrument comfortable and absorbing enough to lose themselves in, while converging independently on the same gap - thrust and strike detection - as the mapping vocabulary's clearest missing piece. Ten participants is a pilot, not a validated usability study, and the claims above should be read accordingly.

The instrument also carries limitations worth naming plainly: Wi-Fi pairing is still a manual, reflash-to-change affair rather than a runtime setting, even after the dropout handling itself was tightened (§2); yaw drift is bounded to sessions of tens of minutes by assumption rather than measurement; the single inertial sensor gives REMORA no sense of a second performer, so it remains a solo instrument for now; and the node-graph editor, while it is what makes the preset catalog reproducible and extensible at all, exposes that same complexity to a performer who only wants to pick a preset and play, with no simplified performance-only view that hides the graph until it is asked for.

One extension is next on the list regardless of the evaluation's findings above: replacing the compiled-in Wi-Fi hotspot join with an ESP-NOW peer-to-peer link between the staff's microcontroller and a second one tethered to the host, which would drop the SSID, password, and destination IP address currently baked into firmware in favour of pairing by MAC address, at the cost of one extra piece of hardware in the signal chain.

7. CONCLUSION

REMORA reads a martial artist's own staff-handling vocabulary - spins, tilts, strikes - directly off a bô and turns it into sound, without baking any single gesture-to-sound behaviour into compiled code: every mapping is instead a small, rewireable patch built from the same handful of source, math, and sink nodes, stretching from the one-line *Simple* mapping to the self-varying *Azimut Kinetic* (§3). Nine presets built from that vocabulary currently ship, pruned down from fifteen once overlapping mappings became apparent (§4). What this paper has established is that the instrument works end to end, from a self-calibrating sensor unit through to three distinct synthesis voices (§2), and that it does so in hands other than its designer's: a ten-participant pilot (§5) found it comfortable and absorbing to play regardless of prior martial-arts or music background, while converging on strike detection as the first thing to fix. Confirming that finding at a larger scale, together with the ESP-NOW rework in §6, is what's next.

8. ACKNOWLEDGMENT

This work was carried out during the first author's internship at CREATE, Aalborg University, Denmark. Firmware, plugin source, enclosure files, and the current preset catalog (plus the retired mappings kept as reference graphs, §4) are available at `https://github.com/TimotheeDvt/REMORA`.

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